PREDICTIVE TASK DURATION ANALYSIS

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PROBLEM STATEMENT

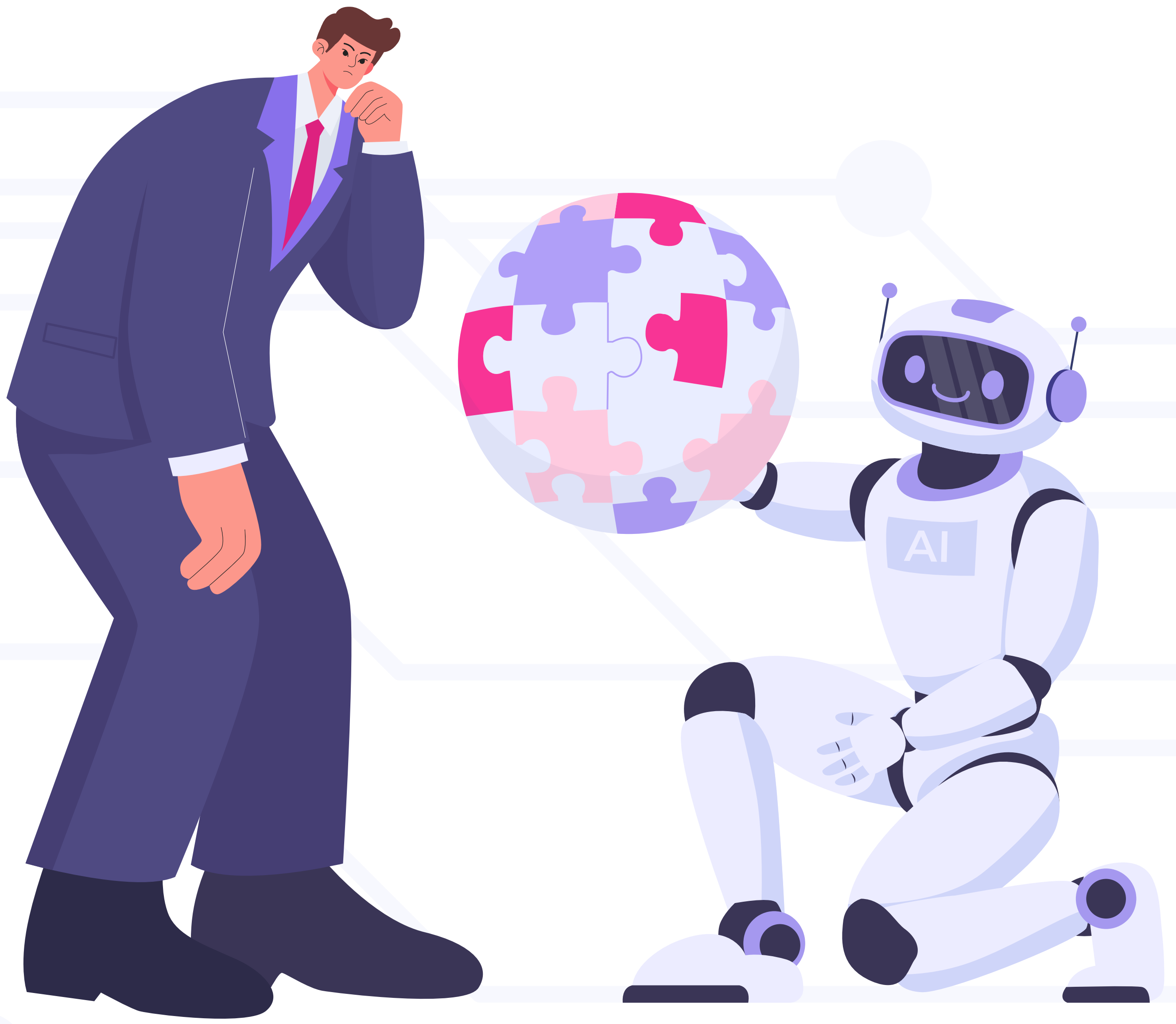
Software development estimation is often subjective, leading to "planning fallacies" and missed deadlines.

OBJECTIVE

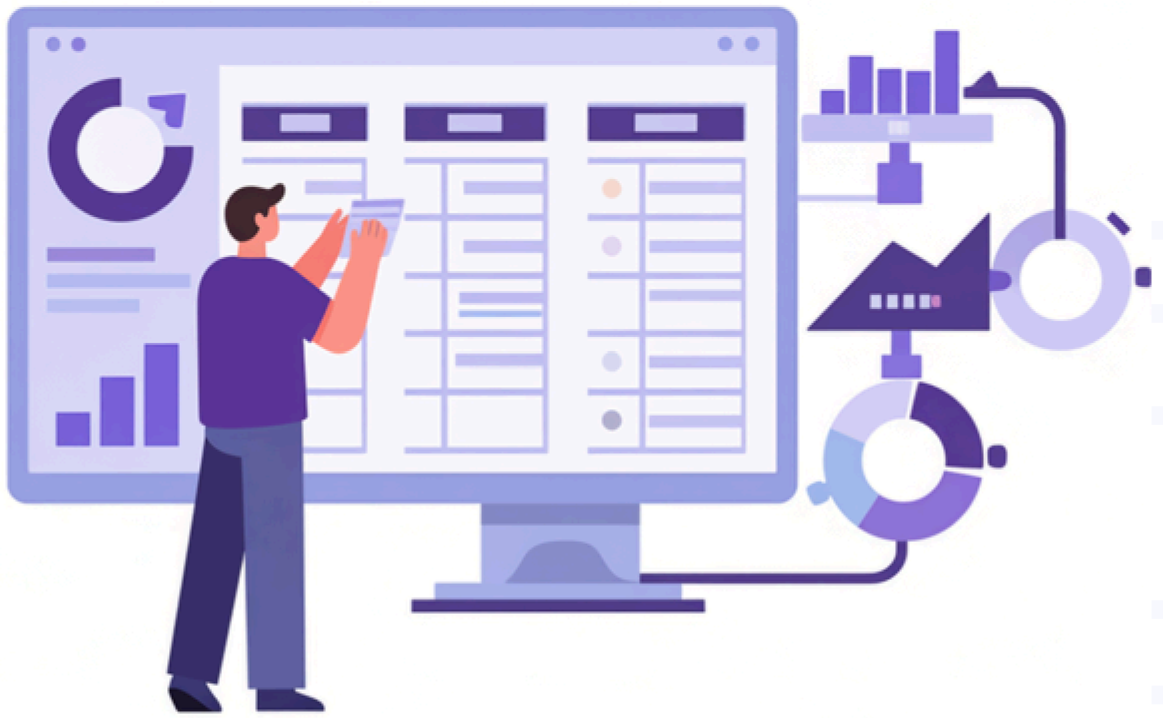
To transition from intuitive estimation to data-driven prediction of task completion times.

RESEARCH QUESTION

Can we accurately forecast Task_Duration_Hours using a combination of technical, workflow, and human-centric data?



TECHNICAL EXECUTION – DATA HANDLING



MODEL INPUTS

- Activity: Coding hours, lines of code, and bugs found.
- Tools: Total AI usage hours.
- Context: Sleep duration, cognitive load, and caffeine intake.

STANDARDIZATION

- Use StandardScaler
- Normalized all features to a common scale.
- Prevents high-value features (Lines of Code) from mathematically dominating low-value features (Coffee Intake).

VALIDATION STRATEGY

- 80/20 Split: 80% for model training; 20% reserved for testing.
- Ensures the model is evaluated on "unseen" data to verify predictive reliability.

LINEAR REGRESSION

BASELINE STRATEGY

Initiated analysis with Linear Regression to establish a performance benchmark.

TECHNICAL IMPLEMENTATION

Utilized OLS method to minimize the sum of squared differences between observed and predicted values

PERFORMANCE METRICS

MAE: 3.00 hours , RMSE: 3.93 hours ,R2: 57%

KEY DRIVERS OF TASK DURATION

THE MAJOR TIME DRIVER

Hours Coding (+4.64): Every hour of active coding is the strongest predictor of total task length.

EFFICIENCY BOOSTERS (TIME SAVERS)

Cognitive Load (-0.28) & Sleep (-0.17)
AI Usage (-0.14)

DURATION INCREASES (DELAY FACTORS)

Lines of Code (+0.24) & Stress (+0.22)
Coffee Intake (+0.21)

PRODUCTIVITY SIGNALS

Commits (-0.13) & Bugs Found (-0.13)

RANDOM FOREST MODELING

THE ENSEMBLE APPROACH

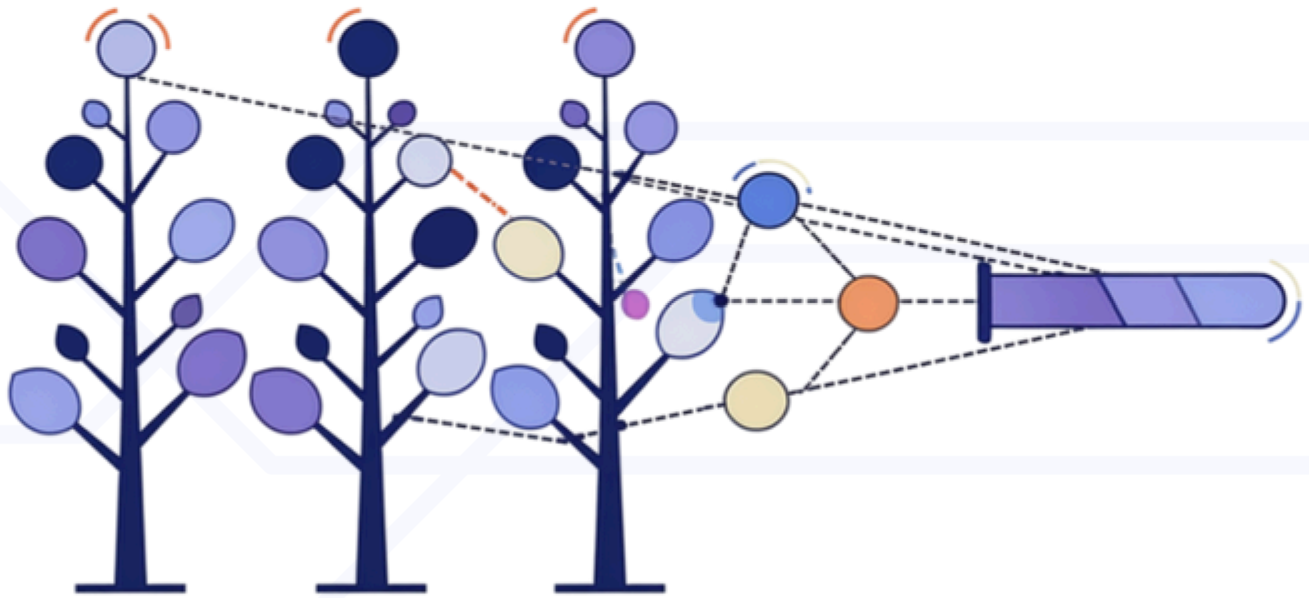
- Utilized a Random Forest Regressor to capture non-linear relationships.
- Built a "forest" of 100 individual decision trees (n_estimators=100) that collaborate to make a final prediction.

PERFORMANCE EVALUATION

- R2: 0.5337
- Mean Absolute Error (MAE): 3.13 hours.

TECHNICAL INSIGHT

- The slight decrease in accuracy compared to Linear Regression (R^2: 0.57) suggests the underlying data patterns are predominantly linear rather than highly complex.



FEATURE IMPORTANCE – KEY DRIVERS

RANKING METHODOLOGY

- Used the feature_importances_ attribute to identify the most influential variables in the Random Forest model.

PRIMARY SIGNAL

- Hours_Coding is the dominant predictor of task duration.

SECONDARY DRIVERS

- AI_Usage_Hours and Stress_Level provide critical context for model accuracy.

ACTIONABLE INSIGHT

- Focus management efforts on these top-tier variables to improve project predictability.

REFINEMENT – FEATURE ENGINEERING & GBR

FEATURE ENGINEERING

- Created the AI-to-Manual Ratio and Total Active Hours to measure AI's relative efficiency.

OPTIMIZATION MODEL

- Implemented Gradient Boosting Regressor (GBR), which builds trees sequentially to minimize errors from previous stages.

OUTCOME

- R2: 0.5186

TECHNICAL CONCLUSION

- Added complexity and custom features did not outperform the baseline, confirming that the primary relationships in this data are robustly linear.

MODEL COMPARISON

LINEAR REGRESSION

- MAE: 3.00h
- R2: 0.5677

RANDOM FOREST

- MAE: 3.00h
- R2: 0.5337

GRADIENT BOOSTING

- MAE: 3.17h
- R2: 0.5186

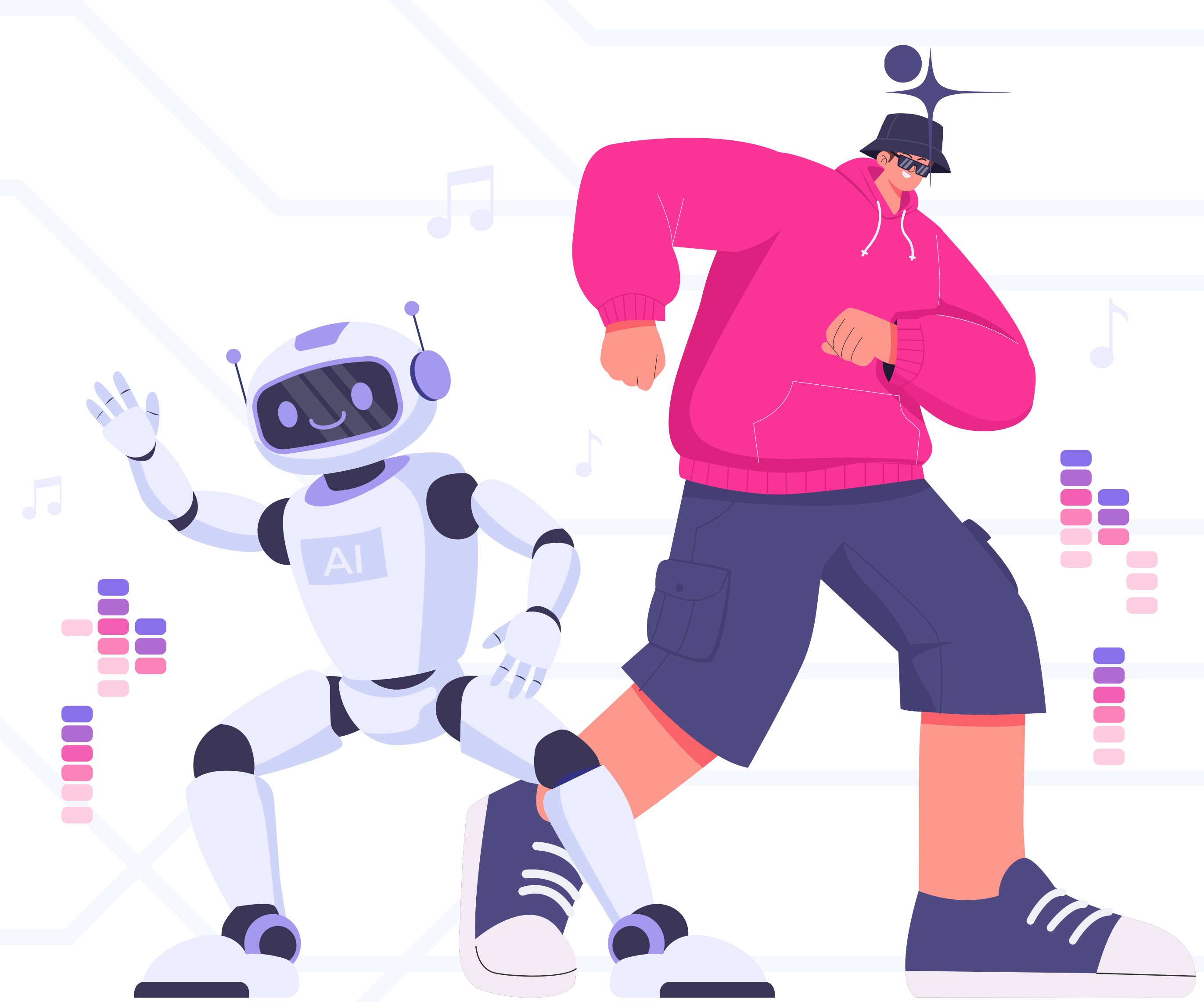
PERFORMANCE SUMMARY

THE "WINNING" MODEL

- Linear Regression provided the lowest error (MAE) and highest explained variance (R2).

KEY TECHNICAL TAKEAWAY

- Increased algorithmic complexity (Ensembles/Boosting) did not improve results, indicating a strong linear relationship between the inputs and task duration.



CONCLUSION & RECOMMENDATIONS

- LINEAR REGRESSION IS THE RECOMMENDED MODEL FOR THIS DATASET, OFFERING THE BEST BALANCE OF SIMPLICITY, TRANSPARENCY, AND PREDICTIVE ACCURACY
- EMPIRICAL EVIDENCE SHOWS THAT AI USAGE HOURS CONSISTENTLY CORRELATE WITH REDUCED TASK DURATION, JUSTIFYING FURTHER INVESTMENT IN AI-ASSISTED WORKFLOWS.
- DEVELOPER WELL-BEING (SLEEP AND STRESS MANAGEMENT) IS A MEASURABLE FACTOR IN DELIVERY SPEED; THESE SHOULD BE INTEGRATED INTO CAPACITY PLANNING.
- PRIORITIZE MANAGING COGNITIVE LOAD AND STRESS LEVELS TO AVOID PROJECT BOTTLENECKS.

THANK YOU!

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